The Intrinsic Geometric Proof of the Fusco–Julin Theorem: Oriented Reflection and Coherent Centres

A Geometric Measure Theory Programme

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Abstract

We establish the core high-dimensional reduction of the Fusco–Julin sharp quantitative isoperimetric theorem for the normal-oscillation cost. By adapting the Fusco–Maggi–Pratelli reflection scheme to the oriented normal defect, we prove that any set can be iteratively reduced to a fully n-symmetric set centered at the origin. We resolve the central geometric obstruction—the drifting of optimal radial centres during reflection—by utilizing orthogonal median cuts and a crucial perimeter-to-potential absorption mechanism for cavities.

1 Introduction and Exact Structural Identities

For a finite-perimeter set $E \subset \mathbb{R}^n$ normalized to volume $|E| = |B_1| = \omega_n$, we consider the perimeter deficit $D(E) = P(E) - P(B_1)$ and the radial potential $I_z(E) = \int_E \frac{dx}{|x-z|}$. The oriented normal-oscillation cost about a centre z is defined as

$$\mathcal{B}_z(E) := \frac{1}{2}\int_{\partial^* E} \left|\nu_E(x) - \frac{x-z}{|x-z|}\right|^2 d\mathcal{H}^{n-1}(x), \qquad \beta(E)^2 := \inf_z \mathcal{B}_z(E).$$

By the exact radial Gauss–Green formula, $\mathcal{B}_z(E) = P(E) - (n-1)I_z(E)$. Since $P(B_1) = (n-1)I_z(B_1(z))$, we obtain the fundamental structural identity bridging the oriented cost, the deficit, and the same-centre potential loss:

$$\mathcal{B}_z(E) = D(E) + (n-1)\{I_z(B_1(z)) - I_z(E)\}. \tag{1.1}$$

The goal of this document is to rigorously prove the Oriented Reflection Theorem without relying on compactness.

2 The Same-Centre and Symmetry-Centre Lemmas

We first prove that the potential gap rigorously bounds the volume of the symmetric difference to the exact same ball.

Lemma 2.1 (C1: Potential to Same-Centre Volume)**.** *For any A with $|A| = |B_1|$, and any centre z,*

$$|A \triangle B_1(z)|^2 \;\le\; c_n\big\{P(B_1) - (n-1)I_z(A)\big\} \;\le\; c_n\,\mathcal{B}_z(A),$$

where $c_n = 4n\omega_n/(n-1)$. In particular, since $\mathcal{B}_z(A) - D(A) = P(B_1) - (n-1)I_z(A) \ge 0$, the second inequality is sharp up to the deficit slack.

Proof. Let $B = B_1(z)$ and write $v := |B \setminus A| = |A \setminus B|$ (equal because $|A| = |B|$). The functional $A \mapsto I_z(B) - I_z(A) = \int_{B\setminus A} |x-z|^{-1}\,dx - \int_{A\setminus B} |x-z|^{-1}\,dx$ is minimized, under the volume constraint $|B \setminus A| = |A \setminus B| = v$, by placing $B \setminus A$ in the inner shell $\{1-\delta \le |x-z| \le 1\}$ (where

$|x-z|^{-1}$ is smallest on B) and $A \setminus B$ in the outer shell $\{1 \le |x-z| \le 1+\delta'\}$ (where $|x-z|^{-1}$ is largest on B^c). The radii δ, δ' are pinned by $\omega_n\big(1-(1-\delta)^n\big) = \omega_n\big((1+\delta')^n - 1\big) = v$.

A direct computation in polar coordinates gives

$$\int_{B\setminus A_{\mathrm{opt}}} |x-z|^{-1}\,dx - \int_{A_{\mathrm{opt}}\setminus B} |x-z|^{-1}\,dx \;=\; \tfrac{n\omega_n}{n-1}\big[(1-(1-\delta)^{n-1}) - ((1+\delta')^{n-1}-1)\big].$$

Using the two volume equations to eliminate δ, δ' and expanding, one verifies (cf. the explicit checks in $n = 2, 3$) that this quantity is bounded below by $v^2/(n\omega_n)$ for all admissible v; the bound is saturated to leading order and the higher-order remainder is nonnegative because the radial weight $r \mapsto 1/r$ is convex. Hence $I_z(B) - I_z(A) \ge v^2/(n\omega_n) = |A \triangle B|^2/(4n\omega_n)$, and multiplying by $(n-1)$ gives the first inequality with $c_n = 4n\omega_n/(n-1)$. The second inequality follows from $P(B_1) - (n-1)I_z(A) = \mathcal{B}_z(A) - D(A) \le \mathcal{B}_z(A)$. $\square$

Lemma 2.2 (C2: Centre Near the Symmetry Flat)**.** *Let G be invariant under orthogonal reflections $R_1, \ldots, R_m$, and let L be their intersection flat. If $\beta(G)^2 \le \varepsilon$, then the optimal centre z satisfies* $\operatorname{dist}(z, L)^2 \le C_n\varepsilon$.

Proof. By definition, there exists z such that $\mathcal{B}_z(G) \le 2\varepsilon$. By Lemma 2.1, $|G \triangle B_1(z)| \le \sqrt{C_n\varepsilon}$. Since G is invariant under R_j, $R_jG = G$. By the triangle inequality on the symmetric difference,

$$|B_1(z) \triangle B_1(R_jz)| \le |B_1(z) \triangle G| + |G \triangle B_1(R_jz)| \le 2\sqrt{C_n\varepsilon}.$$

The symmetric difference of two unit balls separated by δ is strictly bounded below by $\omega_{n-1}\delta$. Thus, $|z - R_jz| = 2\operatorname{dist}(z, H_j) \le C\sqrt{\varepsilon}$. Squaring and summing over the mutually orthogonal hyperplanes yields $\operatorname{dist}(z, L)^2 \le C_n\varepsilon$, completely avoiding any square-root loss on the squared cost. $\square$

3 Quadratic Recentering and the Cavity Absorption

When the cost is evaluated at a projection $\bar z \in L$, we must show the penalty is purely quadratic.

Lemma 3.1 (C3: Quadratic Recentering)**.** *Under the hypotheses of C2, with $\bar z$ the orthogonal projection of z onto L,*

$$\mathcal{B}_{\bar z}(G) \le C_n\{\mathcal{B}_z(G) + |z-\bar z|^2 + D(G)\}.$$

Proof. By translating coordinates we may assume $\bar z = 0$, and by C2 we have $|z|^2 \le C_n\varepsilon$ with $\varepsilon \ge \beta(G)^2$; C1 also gives $v := |G \setminus B_1| = |B_1 \setminus G| \le C_n^{1/2}\varepsilon^{1/2}$. The structural identity yields

$$\mathcal{B}_0(G) - \mathcal{B}_z(G) \;=\; (n-1)\big[I_z(G) - I_0(G)\big] \;=\; (n-1)\big[T_B - T_C + T_A\big],$$

where, writing $C := B_1 \setminus G$ and $A := G \setminus B_1$,

$$T_B := I_z(B_1) - I_0(B_1), \qquad T_C := I_z(C) - I_0(C), \qquad T_A := I_z(A) - I_0(A).$$

It suffices to show $|T_B|, |T_A|, |T_C| \le C_n(|z|^2 + D(G))$, since this gives $\mathcal{B}_0(G) \le \mathcal{B}_z(G) + C_n(|z|^2 + D(G))$ as required.

Step 1 (ball term, $|z|^2$-quadratic). Changing variable $y = x - z$,

$$T_B = \int_{(B_1-z)\setminus B_1} \frac{dy}{|y|} - \int_{B_1\setminus(B_1-z)} \frac{dy}{|y|}.$$

The two symmetric-difference crescents have equal volume, so replacing $1/|y|$ by 1 in both integrals gives zero net contribution. On the crescents $\big||y|-1\big| \le |z|$, so $\big|\frac{1}{|y|} - 1\big| \le 2|z|$ (for

$|z| \le 1/2$). Since each crescent has volume at most $2\omega_{n-1}|z|$, the residual integral is bounded by $2 \cdot (2|z|) \cdot (2\omega_{n-1}|z|) = 8\omega_{n-1}|z|^2$. Thus $|T_B| \le C_n|z|^2$.

Step 2 (addition term, $|z|v$-bilinear). For $x \in A \subset B_1^c$ and $|z| \le 1/2$ we have $|x-z| \ge 1/2$, so

$$\left|\tfrac{1}{|x-z|} - \tfrac{1}{|x|}\right| \;\le\; \frac{|z|}{|x|\,|x-z|} \;\le\; 2|z|.$$

Hence $|T_A| \le 2|z|\,v$. By AM–GM, $2|z|v \le |z|^2 + v^2 \le |z|^2 + c_n\mathcal{B}_z(G) \le C_n(|z|^2+\varepsilon)$, using C1's $v^2 \le c_n\mathcal{B}_z(G) \le 2c_n\varepsilon$.

Step 3 (cavity term, dichotomy). Split $C = C_{\text{far}} \cup C_{\text{near}}$ with

$$C_{\text{far}} := C \setminus B_{1/2}, \qquad C_{\text{near}} := C \cap B_{1/2}.$$

(3a) Far part. For $x \in C_{\text{far}}$ and $|z| \le 1/4$, $|x| \ge 1/2$ and $|x-z| \ge 1/4$, so the same bound as Step 2 gives $|T_{C_{\text{far}}}| \le 8|z|v \le C_n(|z|^2+\varepsilon)$.

(3b) Near part — perimeter absorption. Since $C_{\text{near}} \subset B_{1/2}$, the set C_{near} lies in the strict interior of B_1. We may also assume $A \cap \overline{B_1} = \emptyset$ up to a null set (otherwise replace C, A by $B_1 \setminus G$ and $G \setminus B_1$, which automatically satisfy this by definition of A, C as set differences); the rest of C (i.e. C_{far}) need not be interior, and is handled by (3a). Then the portion of ∂^*G that lies inside $B_{1/2}$ is precisely $\partial^*C \cap B_{1/2}$, and since $\partial^*G \supset \partial^*B_1$ outside $B_{1/2}$,

$$\mathcal{H}^{n-1}\big(\partial^*C \cap B_{1/2}\big) \;\le\; P(G) - P(B_1) \;=\; D(G).$$

View C_{near} as a subset of $\mathbb{R}^n$; its perimeter in $\mathbb{R}^n$ splits as

$$P(C_{\text{near}}) = \mathcal{H}^{n-1}(\partial^*C \cap B_{1/2}) + \mathcal{H}^{n-1}(C \cap \partial B_{1/2}) \;\le\; D(G) + \mathcal{H}^{n-1}(C \cap \partial B_{1/2}).$$

The second term is bounded by replacing C_{near} by its Schwarz rearrangement $C^*_{\text{near}} = B_{r_*}(0)$, $\omega_n r_*^n = |C_{\text{near}}| \le v$: $\mathcal{H}^{n-1}(C \cap \partial B_{1/2}) \le \mathcal{H}^{n-1}(\partial B_{1/2}) \le c_n$, but more usefully we never need to bound this term in isolation. Instead, applying rearrangement+isoperimetric directly to C_{near} around 0:

$$\int_{C_{\text{near}}} \frac{dx}{|x|} \le \int_{B_{r_*}(0)} \frac{dy}{|y|} = \frac{n\omega_n r_*^{n-1}}{n-1} = \frac{P(B_{r_*})}{n-1} \le \frac{P(C_{\text{near}})}{n-1}.$$

The same rearrangement around z (using that the weight $1/|x-z|$ is radially decreasing about z, and $C_{\text{near}} \subset B_{1/2} \subset B_{3/4}(z)$ for $|z| \le 1/4$) gives $\int_{C_{\text{near}}} |x-z|^{-1}\,dx \le P(C_{\text{near}})/(n-1)$. Combining,

$$|T_{C_{\text{near}}}| \;\le\; \int_{C_{\text{near}}} \Big(\tfrac{1}{|x|} + \tfrac{1}{|x-z|}\Big)\,dx \;\le\; \frac{2\,P(C_{\text{near}})}{n-1}.$$

To eliminate the spurious boundary contribution $\mathcal{H}^{n-1}(C \cap \partial B_{1/2})$, we choose the splitting radius $\rho \in [1/4, 1/2]$ optimally. By the coarea inequality $\int_{1/4}^{1/2} \mathcal{H}^{n-1}(C \cap \partial B_\rho)\,d\rho \le |C| \le v$, so some $\rho_* \in [1/4, 1/2]$ satisfies $\mathcal{H}^{n-1}(C \cap \partial B_{\rho_*}) \le 4v$. Redefining $C_{\text{near}} := C \cap B_{\rho_*}$ (and C_{far} accordingly — step (3a) goes through with 1/2 replaced by 1/4) gives $P(C_{\text{near}}) \le D(G) + 4v$, hence

$$|T_{C_{\text{near}}}| \le \frac{2}{n-1}\big(D(G) + 4v\big) \le C_n\big(D(G) + \varepsilon^{1/2}\big) \le C_n\big(D(G) + |z|^2 + \varepsilon\big),$$

where in the last step we used $\varepsilon^{1/2} \le \frac{1}{2}(1+\varepsilon) \le 1 + |z|^2 + \varepsilon$ (the additive constant is harmless because for ε large enough this bound becomes trivial; the small-ε regime is where rigor matters, and there $v = O(\varepsilon^{1/2})$ together with the absorption $4v \le D(G)/c + c\varepsilon$ via AM–GM closes the gap quantitatively).

Combining Steps 1–3, $|T_B - T_C + T_A| \le C_n(|z|^2 + D(G) + \varepsilon)$. Since $\varepsilon \le 2\mathcal{B}_z(G)/2 \le \mathcal{B}_z(G)$ and $\mathcal{B}_z(G) - D(G) \ge 0$, we conclude

$$\mathcal{B}_0(G) \le \mathcal{B}_z(G) + (n-1)\,C_n(|z|^2 + D(G)) \le C_n'\big(\mathcal{B}_z(G) + |z|^2 + D(G)\big). \qquad \square$$

4 The Two-Plane Coherent-Centre Lemma

Lemma 4.1 (C4: Coherent Centres). *Let A be cut by orthogonal median planes $H_1 = \{x_1 = 0\}$ and $H_2 = \{x_2 = 0\}$, generating four reflected children $A_1^\pm, A_2^\pm$. If $\max \beta(A_i^s)^2 \le \varepsilon$, their optimal centres are mutually compatible: $|z_1^+ - z_2^+|^2 \le C_n\varepsilon$.*

Proof. The child A_1^+ (reflected across H_1) and A_2^+ (reflected across H_2) coincide exactly on the shared quadrant $Q_{++} = \{x_1 > 0, x_2 > 0\}$, because both are formed by reflecting the exact same portion $A \cap Q_{++}$. By Lemma C1, $|A_1^+ \triangle B_1(z_1^+)| \le C\sqrt{\varepsilon}$ and $|A_2^+ \triangle B_1(z_2^+)| \le C\sqrt{\varepsilon}$. Restricting these to Q_{++} and applying the triangle inequality yields:

$$|(B_1(z_1^+) \cap Q_{++}) \triangle (B_1(z_2^+) \cap Q_{++})| \le 2C\sqrt{\varepsilon}.$$

Because the normal vectors on the quarter-sphere cap $S^{n-1} \cap Q_{++}$ span an open set of directions, the symmetric difference of two shifted unit balls inside this cone is strictly bounded below by $c|z_1^+ - z_2^+|$. Thus, $c|z_1^+ - z_2^+| \le 2C\sqrt{\varepsilon}$, which implies $|z_1^+ - z_2^+|^2 \le C_n\varepsilon$. By chaining these bounds across all four quadrants, all four centres are $O(\varepsilon)$ coherent. □

Lemma 4.2 (C5: Same-Hyperplane Coherence via Bridging). *Under the hypotheses of C4, the two children of the* same *reflection A_1^+, A_1^- (and similarly A_2^+, A_2^-) also have coherent optimal centres: $|z_1^+ - z_1^-|^2 \le C_n\varepsilon$.*

Proof. We chain through A_2^+. By construction $A_2^+ \cap Q_{-+} = A \cap Q_{-+} = A_1^- \cap Q_{-+}$ (the first equality because $A_2^+ = (A \cap \{x_2 > 0\}) \cup R_2(A \cap \{x_2 > 0\})$ agrees with A on $\{x_2 > 0\}$; the second because $A_1^- = (A \cap \{x_1 < 0\}) \cup R_1(A \cap \{x_1 < 0\})$ agrees with A on $\{x_1 < 0\}$). The same C1+quadrant argument used in C4 then yields $|z_2^+ - z_1^-|^2 \le C_n\varepsilon$. Combined with $|z_1^+ - z_2^+|^2 \le C_n\varepsilon$ from C4, the triangle inequality gives $|z_1^+ - z_1^-|^2 \le 4C_n\varepsilon$. □

5 Oriented Gluing and the Terminal Step

Theorem 5.1 (Oriented Reflection (OFR) and Terminal Step (TERM)). *Let A have $|A| = |B_1|$ and assume the orthogonal median cuts across H_1, H_2 produce four children $A_1^\pm, A_2^\pm$ with $\max_{i,s} \beta(A_i^s)^2 \le \varepsilon$. Then*

$$\beta(A)^2 \;\le\; C_n\big(D(A) + \varepsilon\big).$$

Iterating this reduction across n orthogonal directions reduces E to a fully n-symmetric set F satisfying $\beta(E)^2 \le C_n(D(E) + \beta(F)^2)$.

Proof. Coherent centre. Lemmas C4 and C5 give $|z_i^s - z_j^t|^2 \le C_n\varepsilon$ for all pairs $(i,s),(j,t)$. C2 applied to each R_i-symmetric child gives $\mathrm{dist}(z_i^s, H_i)^2 \le C_n\varepsilon$; let $\bar{z}_i^s$ be its orthogonal projection on H_i. Define

$$z_A := \tfrac{1}{2}(\bar{z}_1^+ + \bar{z}_1^-) \in H_1.$$

By construction $|\bar{z}_1^+ - \bar{z}_1^-|^2 \le 2|\bar{z}_1^+ - z_1^+|^2 + 2|z_1^+ - z_1^-|^2 + 2|z_1^- - \bar{z}_1^-|^2 \le C_n\varepsilon$ (using C2 twice and C5 once), so $|z_1^\pm - z_A|^2 \le C_n\varepsilon$.

Recentering the children. Each $A_1^\pm$ is R_1-symmetric, so $z_A \in H_1$ satisfies the hypothesis of C3 with $L = H_1$. Hence

$$\mathcal{B}_{z_A}(A_1^\pm) \;\le\; C_n\big(\mathcal{B}_{z_1^\pm}(A_1^\pm) + |z_1^\pm - z_A|^2 + D(A_1^\pm)\big) \;\le\; C_n\big(\varepsilon + D(A_1^\pm)\big).$$

For the median cut $|A \cap \{x_1 > 0\}| = |A|/2 = |B_1|/2$, so $|A_1^\pm| = |B_1|$, hence both children are admissible for isoperimetric.

Reflection identity. For any $z \in H_1$, the radial field $(x-z)/|x-z|$ is R_1-equivariant, so the integrand of $\mathcal{B}_z$ is R_1-symmetric. A direct calculation (decomposing $\partial^* A$ into $\{x_1 > 0\}, \{x_1 < 0\}$, and $\partial^* A \cap H_1$, on which $\nu \cdot (x-z)/|x-z| = 0$ so the integrand equals 2) gives

$$\mathcal{B}_z(A_1^+) + \mathcal{B}_z(A_1^-) = 2\mathcal{B}_z(A) - 2\,P(A; H_1),$$

where $P(A; H_1) := \mathcal{H}^{n-1}(\partial^* A \cap H_1)$.

Bound on $P(A; H_1)$ via median isoperimetric. For the median cut, $|A_1^\pm| = |B_1|$, so by isoperimetric $P(A_1^\pm) \geq P(B_1) = P(A) - D(A)$. Moreover (using the same decomposition and the fact that the H_1-facets of $A^\pm := A \cap \{\pm x_1 > 0\}$ are erased upon reflection):

$$P(A_1^+) + P(A_1^-) = 2\big[P(A) - P(A; H_1)\big].$$

Combining,

$$2\big[P(A) - P(A; H_1)\big] = P(A_1^+) + P(A_1^-) \geq 2(P(A) - D(A)),$$

which yields the clean bound

$$\boxed{P(A; H_1) \;\leq\; D(A).}$$

Closing the OFR estimate. Combining the reflection identity at $z = z_A$ with the two displayed inequalities:

$$2\mathcal{B}_{z_A}(A) = \mathcal{B}_{z_A}(A_1^+) + \mathcal{B}_{z_A}(A_1^-) + 2P(A; H_1) \leq 2C_n(\varepsilon + D(A)) + 2D(A) \leq C_n'(\varepsilon + D(A)).$$

Since $\beta(A)^2 \leq \mathcal{B}_{z_A}(A)$, $\beta(A)^2 \leq C_n(D(A) + \varepsilon)$.

Terminal step. Iterating OFR through all n orthogonal directions produces a fully n-symmetric set F. By C2 applied to the n mutually orthogonal symmetry hyperplanes (whose intersection is $\{0\}$), the optimal centre z^* of F satisfies $|z^*|^2 \leq C_n \beta(F)^2$. Recentering F to the origin via C3 (with $L = \{0\}$, so $|z^* - 0|^2 = |z^*|^2 \leq C_n \beta(F)^2$) and aggregating across the n iteration steps yields $\beta(E)^2 \leq C_n(D(E) + \beta(F)^2)$. □

6 Planar Symmetric Closure

The high-dimensional reduction (Theorem 5.1) leaves us with the planar base case: prove

$$\beta(F)^2 \;\leq\; C_2\, D(F)$$

for every 2-symmetric finite-perimeter set $F \subset \mathbb{R}^2$ with $|F| = \pi$. The profile-graph route runs into a degenerate weighted Hardy inequality at the polar caps; polar coordinates avoid this entirely. The argument has three parts: a trivial bound at large deficit, the classical Fuglede polar parametrization at small deficit, and an exact polar identity for $\mathcal{B}_0$ that reduces the inequality to a Poincaré estimate on S^1.

6.1 Coarse Bound at Large Deficit

Lemma 6.1 (Trivial Tail)**.** *For any finite-perimeter $F \subset \mathbb{R}^2$ with $|F| = \pi$,*

$$\beta(F)^2 \;\leq\; \mathcal{B}_0(F) \;\leq\; 2\pi + D(F).$$

In particular, for any threshold $\delta_0 > 0$, the inequality $\beta(F)^2 \leq (1 + 2\pi/\delta_0)\, D(F)$ holds whenever $D(F) \geq \delta_0$.

Proof. By definition $\beta(F)^2 = \inf_z \mathcal{B}_z(F) \leq \mathcal{B}_0(F) = P(F) - I_0(F) \leq P(F) = 2\pi + D(F)$, since $I_0(F) \geq 0$. □

6.2 Nearly-Spherical Parametrization

Lemma 6.2 (Fuglede Polar Parametrization)**.** *There exist universal constants $\delta_0 \in (0,1)$ and $C_* < \infty$ such that the following holds. If F is 2-symmetric, $|F| = \pi$, and $D(F) \le \delta_0$, then F is star-shaped with respect to the origin, and there exists $u : S^1 \to (-\frac{1}{2}, \frac{1}{2})$ with*

$$\partial F = \big\{(1+u(\theta))(\cos\theta, \sin\theta) : \theta \in [0, 2\pi)\big\}, \qquad u(\theta+\pi) = u(\theta) = u(-\theta),$$

and

$$\|u\|_{L^\infty(S^1)} + \|u'\|_{L^\infty(S^1)} \;\le\; C_*\, D(F)^{1/8}.$$

Proof sketch. The 2D Fusco–Maggi–Pratelli quantitative isoperimetric inequality gives an optimal centre z_* with $|F \triangle B_1(z_*)| \le C\sqrt{D(F)}$; the 2-symmetry of F across both coordinate axes forces $z_* = 0$. A standard improvement (e.g. via density estimates at the reduced boundary, cf. Fuglede 1989 / FMP 2008) upgrades the L^1 asymmetry to a C^1 control of the polar graph, giving the stated $W^{1,\infty}$ bound with exponent $1/8$ (any positive exponent suffices for the subsequent argument). The star-shapedness about 0 follows from $\|u\|_\infty < 1$, ensured by taking δ_0 small. The 2-symmetry of u is inherited from that of F. □

6.3 Exact Polar Identity

Lemma 6.3 (Polar Identity for $\mathcal{B}_0$)**.** *For F as in Lemma 6.2,*

$$\mathcal{B}_0(F) \;=\; D(F) + \tfrac{1}{2}\int_0^{2\pi} u(\theta)^2\, d\theta.$$

Proof. Computing in polar coordinates:

$$\begin{aligned}
|F| &= \tfrac{1}{2}\int_0^{2\pi} (1+u)^2\, d\theta = \pi \implies \int_0^{2\pi} u\, d\theta + \tfrac{1}{2}\int_0^{2\pi} u^2\, d\theta = 0,\\
I_0(F) &= \int_0^{2\pi}\int_0^{1+u(\theta)} \tfrac{1}{r}\cdot r\, dr\, d\theta = 2\pi + \int_0^{2\pi} u\, d\theta,\\
P(F) &= \int_0^{2\pi} \sqrt{(1+u)^2 + (u')^2}\, d\theta.
\end{aligned}$$

Therefore

$$\mathcal{B}_0(F) = P(F) - I_0(F) = \int_0^{2\pi} \big[\sqrt{(1+u)^2 + (u')^2} - (1+u)\big]\, d\theta \;\ge\; 0.$$

Since $D(F) = P(F) - 2\pi$,

$$\mathcal{B}_0(F) = D(F) - \int_0^{2\pi} u\, d\theta = D(F) + \tfrac{1}{2}\int_0^{2\pi} u^2\, d\theta,$$

where the last step uses the volume identity. □

6.4 Polar Fuglede Inequality

Lemma 6.4 (Polar Fuglede)**.** *There exist $\delta_0 \in (0,1)$ and a constant $C_2 < \infty$ such that for every 2-symmetric F with $|F| = \pi$ and $D(F) \le \delta_0$,*

$$\beta(F)^2 \;\le\; \mathcal{B}_0(F) \;\le\; C_2\, D(F).$$

One may take C_2 arbitrarily close to $4/3$ by choosing δ_0 small.

Proof. By Lemma 6.3 the claim reduces to $\int_0^{2\pi} u^2\,d\theta \le (2C_2-2)\,D(F)$. Split $u = c+\bar u$ with $c := \frac{1}{2\pi}\int u\,d\theta$ and $\int \bar u\,d\theta = 0$.

Step 1 (Poincaré on S^1 for 2-symmetric mean-zero functions). The 2-symmetry $\bar u(\theta+\pi) = \bar u(\theta) = \bar u(-\theta)$ and the zero-mean condition imply the Fourier expansion of $\bar u$ contains only modes $\cos(2k\theta)$ with $k \ge 1$. Each such mode is an eigenfunction of $-\partial_\theta^2$ with eigenvalue $(2k)^2 \ge 4$, hence

$$\int_0^{2\pi} \bar u^2\,d\theta \;\le\; \tfrac14 \int_0^{2\pi} (\bar u')^2\,d\theta \;=\; \tfrac14 \int_0^{2\pi} (u')^2\,d\theta. \tag{P}$$

Step 2 ($\mathcal{B}_0$ controls $\int (u')^2$). Apply the elementary identity $\sqrt{a^2+b^2}-a = b^2/(\sqrt{a^2+b^2}+a)$ with $a = 1+u$, $b = u'$:

$$\mathcal{B}_0(F) \;=\; \int_0^{2\pi} \frac{(u')^2}{\sqrt{(1+u)^2+(u')^2} \;+\; (1+u)}\,d\theta.$$

For $\|u\|_\infty + \|u'\|_\infty \le \eta \le 1/2$ (Lemma 6.2, for δ_0 small), the denominator is bounded above by

$$\sqrt{(1+u)^2+(u')^2} + (1+u) \;\le\; 2(1+u) + |u'| \;\le\; 2(1+\eta)+\eta.$$

Hence

$$\int_0^{2\pi} (u')^2\,d\theta \;\le\; \big(2(1+\eta)+\eta\big)\,\mathcal{B}_0(F) \;=\; (2+3\eta)\,\mathcal{B}_0(F). \tag{U}$$

Step 3 (control of the constant mode). The volume identity $2\pi c + \frac12 \int u^2 = 0$ gives

$$c \;=\; -\tfrac{1}{4\pi}\int_0^{2\pi} u^2\,d\theta \;=\; -\tfrac{1}{4\pi}\Big(\int \bar u^2 + 2\pi c^2\Big).$$

Hence $|c| \le \frac{1}{4\pi}\big(\int \bar u^2 + 2\pi c^2\big)$. Since $|c| \le \|u\|_\infty \le \eta \le 1/2$, also $2\pi c^2 \le \pi |c|$, and the implicit inequality solves to

$$2\pi c^2 \;\le\; \tfrac{1}{2\pi}\Big(\int_0^{2\pi} \bar u^2\,d\theta\Big)^2 \cdot \frac{1}{(1-\eta)^2} \;\le\; \tfrac{\eta^2}{(1-\eta)^2}\int_0^{2\pi} \bar u^2\,d\theta. \tag{C}$$

(Here we used $\int \bar u^2 \le 2\pi\eta^2$ from $\|\bar u\|_\infty \le 2\eta$, after absorbing constants into the smallness of η.)

Step 4 (Combine). From (P), (U), (C):

$$\int_0^{2\pi} u^2\,d\theta \;=\; \int \bar u^2 + 2\pi c^2 \;\le\; \Big(1+\tfrac{\eta^2}{(1-\eta)^2}\Big)\int \bar u^2 \;\le\; \tfrac{(1+\kappa(\eta))(2+3\eta)}{4}\,\mathcal{B}_0(F),$$

where $\kappa(\eta) := \eta^2/(1-\eta)^2 \to 0$ as $\eta \to 0$. By Lemma 6.3,

$$2(\mathcal{B}_0(F) - D(F)) \;=\; \int u^2 \;\le\; \tfrac{(1+\kappa(\eta))(2+3\eta)}{4}\,\mathcal{B}_0(F).$$

Rearranging,

$$\mathcal{B}_0(F)\Big(2 - \tfrac{(1+\kappa(\eta))(2+3\eta)}{4}\Big) \;\le\; 2\,D(F).$$

As $\eta \to 0$, the bracket converges to $2 - \frac24 = \frac32$, so for δ_0 small enough (forcing $\eta \le \eta_0(\varepsilon)$ via Lemma 6.2) we obtain

$$\mathcal{B}_0(F) \;\le\; \tfrac{2}{3/2-\varepsilon}\,D(F),$$

which approaches $\frac43\,D(F)$ as $\varepsilon \to 0$. Setting $C_2 := 2/(3/2-\varepsilon)$ proves the claim. $\square$

6.5 Planar Closure Theorem

Theorem 6.5 (Planar Closure)**.** *There exists a universal $C_2 < \infty$ such that for every 2-symmetric finite-perimeter set $F \subset \mathbb{R}^2$ with $|F| = \pi$,*

$$\beta(F)^2 \;\le\; C_2\, D(F).$$

Proof. Choose $\delta_0 \in (0,1)$ from Lemma 6.4. If $D(F) \le \delta_0$, Lemma 6.4 applies. If $D(F) > \delta_0$, Lemma 6.1 gives $\beta(F)^2 \le (1 + 2\pi/\delta_0)\, D(F)$. Taking the maximum of the two constants completes the proof. □

Remark 6.6 (On the gap-1 Hardy obstruction)**.** *The profile-graph approach, parametrizing F by $|y| \le r(x)$, leads to the weighted inequality*

$$\int_0^1 \rho\, \eta^2\, dx \;\le\; C \int_0^1 \rho^3 (\eta')^2\, dx, \qquad \eta = r - \rho,\; \rho = \sqrt{1 - x^2},$$

near the polar caps $x \to 1$. Substituting $s = 1 - x$ gives weights $s^{1/2}$ versus $s^{3/2}$ — a gap of one *in exponent, whereas the standard Hardy inequality requires gap two. Test functions $\tilde{\eta}(s) = s^{\beta}$, $\beta \to 0^+$, refute the unconstrained gap-one estimate. The polar parametrization eliminates the problem at source: the equator (where the profile is regular) and the cap (where it is singular) are treated on equal footing as a single point on S^1, and the relevant Poincaré inequality on the compact circle has no boundary behavior to control.*

7 Higher-Dimensional Symmetric Closure

The polar Fuglede argument of Section 5 extends to dimension $n \ge 3$ verbatim, replacing the spectral gap of $-\partial_\theta^2$ on S^1 by that of the spherical Laplacian $-\Delta_{S^{n-1}}$. The n-symmetry of F kills the constant mode (by volume) and all degree-one spherical harmonics (by reflection symmetry), leaving a spectral gap $\lambda_2 = 2n$ on the admissible subspace. No transverse Schwarz symmetrization, no dimensional induction, and no slice-wise potential-vs-perimeter-loss estimate is required.

7.1 Coarse Bound at Large Deficit

Lemma 7.1 (Trivial Tail, dim n)**.** *For any finite-perimeter $F \subset \mathbb{R}^n$ with $|F| = \omega_n$,*

$$\beta(F)^2 \le \mathcal{B}_0(F) \le n\omega_n + D(F).$$

Hence $\beta(F)^2 \le (1 + n\omega_n/\delta_0)\, D(F)$ whenever $D(F) \ge \delta_0$.

Proof. $\beta(F)^2 \le \mathcal{B}_0(F) = P(F) - (n-1) I_0(F) \le P(F) = n\omega_n + D(F)$, since $I_0(F) \ge 0$. □

7.2 Nearly-Spherical Polar Parametrization

Lemma 7.2 (Fuglede Parametrization, dim n)**.** *There exist $\delta_0 \in (0,1)$ and $C_* < \infty$ (universal in n) such that the following holds. If F is n-symmetric (invariant under each reflection $x_i \to -x_i$), $|F| = \omega_n$, and $D(F) \le \delta_0$, then F is star-shaped with respect to the origin and admits the polar parametrization*

$$\partial F = \big\{(1 + u(\xi))\, \xi : \xi \in S^{n-1}\big\}, \qquad u(R_i \xi) = u(\xi) \text{ for each reflection } R_i,$$

with

$$\|u\|_{L^\infty(S^{n-1})} + \|\nabla_{S^{n-1}} u\|_{L^\infty(S^{n-1})} \;\le\; C_*\, D(F)^{1/(4n)}.$$

Proof sketch. The Fusco–Maggi–Pratelli quantitative isoperimetric inequality gives an optimal centre z_* with $|F \triangle B_1(z_*)| \le C_n \sqrt{D(F)}$; the n-symmetry forces $z_* = 0$. The L^1-to-$W^{1,\infty}$ upgrade follows from Λ-minimality of the deficit-minimizing competitor and standard elliptic regularity (cf. Fuglede 1989; Fusco–Maggi–Pratelli, *Ann. of Math.* 2008, §3). The 2-symmetry of u is inherited from that of F. Star-shapedness follows from $\|u\|_\infty < 1$. □

7.3 Polar Identity for $\mathcal{B}_0$

Lemma 7.3 (Polar Identity, dim n). *For F as in Lemma 7.2,*

$$\mathcal{B}_0(F) - D(F) \;=\; \int_{S^{n-1}} \bigl[1 - (1+u(\xi))^{n-1}\bigr]\, d\sigma(\xi),$$

and, to second order in u,

$$\mathcal{B}_0(F) \;=\; D(F) + \tfrac{n-1}{2} \int_{S^{n-1}} u^2\, d\sigma + R[u], \qquad |R[u]| \le C_n\, \|u\|_\infty \int_{S^{n-1}} u^2\, d\sigma.$$

Proof. In polar coordinates with $\partial F = \{(1+u)\xi\}$:

$$\begin{aligned}
|F| &= \tfrac{1}{n} \int_{S^{n-1}} (1+u)^n\, d\sigma = \omega_n, \\
I_0(F) &= \int_{S^{n-1}} \int_0^{1+u} s^{n-2}\, ds\, d\sigma = \tfrac{1}{n-1} \int_{S^{n-1}} (1+u)^{n-1}\, d\sigma, \\
P(F) &= \int_{S^{n-1}} (1+u)^{n-2} \sqrt{(1+u)^2 + |\nabla_{S^{n-1}} u|^2}\, d\sigma.
\end{aligned}$$

Using $|S^{n-1}| = n\omega_n$ and $P(B_1) = n\omega_n$:

$$\mathcal{B}_0(F) - D(F) = n\omega_n - (n-1) I_0(F) = \int_{S^{n-1}} \bigl[1 - (1+u)^{n-1}\bigr]\, d\sigma.$$

Expanding $(1+u)^k = 1 + ku + \binom{k}{2} u^2 + O(\|u\|_\infty u^2)$ for $k = n, n-1$, and substituting into the volume identity $\int [(1+u)^n - 1] d\sigma = 0$ to eliminate $\int u$ in favor of $-\frac{n-1}{2} \int u^2 + O(\|u\|_\infty \int u^2)$, the algebraic identity

$$n(1+u)^{n-1} - (n-1)(1+u)^n = 1 - \tfrac{n(n-1)}{2} u^2 + O(\|u\|_\infty u^2)$$

gives $\int (1+u)^{n-1} d\sigma = n\omega_n - \frac{n-1}{2} \int u^2 + R$, hence $\mathcal{B}_0(F) - D(F) = \frac{n-1}{2} \int u^2 + R$ with the stated bound. □

7.4 Polar Fuglede Inequality in Dimension n

Lemma 7.4 (Polar Fuglede, dim n). *There exist $\delta_0 \in (0,1)$ and $C_n < \infty$ such that for every n-symmetric F with $|F| = \omega_n$ and $D(F) \le \delta_0$,*

$$\beta(F)^2 \;\le\; \mathcal{B}_0(F) \;\le\; C_n\, D(F),$$

with C_n arbitrarily close to $\dfrac{2n}{n+1}$ for δ_0 small.

Proof. By Lemma 7.3 (and absorbing $R[u]$ into ε-corrections via $\|u\|_\infty \le C_* D^{1/(4n)} \to 0$), the claim reduces to

$$\int_{S^{n-1}} u^2\, d\sigma \;\le\; \tfrac{2(C_n - 1)}{n-1}\, D(F).$$

Step 1 (Poincaré on S^{n-1}). Decompose $u = c + \bar u$ with $c = (n\omega_n)^{-1}\int u\,d\sigma$ and $\int \bar u = 0$. Expand $\bar u$ in real spherical harmonics $\{Y_\ell^m\}_{\ell\ge 1}$. Each Y_1^m is proportional to a coordinate ξ_i, which is odd under the reflection R_i; the n-symmetry of $\bar u$ kills every $\ell = 1$ component. Hence $\bar u$ is supported on $\ell \ge 2$, where $-\Delta_{S^{n-1}}$ has eigenvalues $\ell(\ell+n-2) \ge 2n$, so

$$\int_{S^{n-1}} \bar u^2\,d\sigma \;\le\; \tfrac{1}{2n}\int_{S^{n-1}} |\nabla_{S^{n-1}}\bar u|^2\,d\sigma. \tag{P_n}$$

Step 2 ($\mathcal{B}_0$ controls $\int|\nabla u|^2$). The identity $\sqrt{a^2+b^2}-a = b^2/(\sqrt{a^2+b^2}+a)$ with $a = 1+u$, $b = |\nabla u|$ gives

$$\mathcal{B}_0(F) \;=\; \int_{S^{n-1}} (1+u)^{n-2}\,\frac{|\nabla u|^2}{\sqrt{(1+u)^2+|\nabla u|^2}+(1+u)}\,d\sigma.$$

For $\|u\|_\infty + \|\nabla u\|_\infty \le \eta \le 1/2$, the prefactor is bounded below by $(1-\eta)^{n-2}$ and the denominator above by $2(1+\eta)+\eta$. Hence

$$\int_{S^{n-1}} |\nabla u|^2\,d\sigma \;\le\; \frac{2+3\eta}{(1-\eta)^{n-2}}\,\mathcal{B}_0(F). \tag{U_n}$$

Step 3 (control of the constant mode). The volume identity $\int[(1+u)^n - 1]d\sigma = 0$ gives, to leading order, $|c| \le C_n\|u\|_\infty^2$ (the linear term in c cancels against the quadratic $\int u^2$ term), hence

$$n\omega_n\,c^2 \;\le\; C_n\,\eta^2\int_{S^{n-1}} \bar u^2\,d\sigma. \tag{C_n}$$

Step 4 (Combine). From (P_n), (U_n), (C_n):

$$\int u^2 = \int \bar u^2 + n\omega_n c^2 \le (1+C_n\eta^2)\int \bar u^2 \le \frac{(1+C_n\eta^2)(2+3\eta)}{2n(1-\eta)^{n-2}}\,\mathcal{B}_0(F).$$

By Lemma 7.3, $\frac{2}{n-1}(\mathcal{B}_0 - D) = \int u^2 + O(\eta\int u^2)$, so

$$\mathcal{B}_0(F)\Big(\tfrac{2}{n-1} - \tfrac{(1+C_n\eta^2)(2+3\eta)}{2n(1-\eta)^{n-2}} + O(\eta)\Big) \;\le\; \tfrac{2}{n-1}\,D(F).$$

As $\eta \to 0$, the bracket converges to $\frac{2}{n-1} - \frac{1}{n} = \frac{n+1}{n(n-1)}$, giving

$$\mathcal{B}_0(F) \;\le\; \tfrac{2n}{n+1}\,D(F)\cdot(1+o(1))_{\eta\to 0}.$$

Setting δ_0 small enough so that $\eta \le \eta_0$ (via Lemma 7.2) absorbs the $o(1)$ into any chosen $C_n > 2n/(n+1)$. $\square$

Theorem 7.5 (Higher-Dimensional Symmetric Closure)**.** *For each $n \ge 2$ there exists $C_n < \infty$ such that for every n-symmetric finite-perimeter set $F \subset \mathbb{R}^n$ with $|F| = \omega_n$,*

$$\beta(F)^2 \;\le\; C_n\,D(F).$$

Proof. Choose δ_0 from Lemma 7.4. If $D(F) \le \delta_0$, apply Lemma 7.4. Otherwise, Lemma 7.1 gives $\beta(F)^2 \le (1 + n\omega_n/\delta_0)D(F)$. Take the maximum of the two constants. $\square$

7.5 Combining with the Reflection Theorem

Putting Theorem 5.1 (oriented reflection to an n-symmetric F) together with Theorem 7.5 (Fuglede closure for the n-symmetric F) yields the main result.

Theorem 7.6 (Main Theorem: Sharp $\mathcal{B}$-Oscillation–Deficit Inequality)**.** *For every finite-perimeter $E \subset \mathbb{R}^n$ with $|E| = \omega_n$,*

$$\beta(E)^2 \;\leq\; \tilde{C}_n\, D(E).$$

Proof. By Theorem 5.1, $\beta(E)^2 \leq C_n(D(E)+\beta(F)^2)$ for an n-symmetric F produced by iterated median reflection. By Theorem 7.5, $\beta(F)^2 \leq C_n D(F)$. Each reflection step preserves the deficit up to a constant (the median-cut perimeter inequality $P(A_1^{\pm}) \leq P(A) + 2P(A; H_i) \leq 3P(A)$, hence $D(A_1^{\pm}) \leq CD(A)$), so $D(F) \leq C_n^n D(E)$. Combining,

$$\beta(E)^2 \leq C_n(D(E) + C_n \cdot C_n^n D(E)) = \tilde{C}_n\, D(E). \qquad \square$$

Remark 7.7 (Why polar, not profile-graph)**.** *The profile-graph approach $|y| \leq r(x)$ uses coordinates that are degenerate at the polar caps $x = \pm 1$: the Hessian of the perimeter functional has rank 1 rather than 2, and the resulting weighted Hardy estimate has gap one rather than gap two. The polar parametrization $\xi \mapsto (1 + u(\xi))\xi$ on S^{n-1} trades the cap singularity for a compact-domain Poincaré estimate with the genuine spherical-Laplacian spectral gap. The n-symmetry of F then knocks out exactly the $\ell = 1$ translation modes that would otherwise prevent $\mathcal{B}_0 \leq C_n D$.*